Thirty Years of Superconducting Materials Research, 1996–2026: A Scoping Review Based on a Provenance-Traced LLM-Curated arXiv Materials Database

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Abstract

Superconducting materials research over the last three decades has not evolved as a smooth monotonic progression toward higher critical temperature. Using SCLib, a provenance-traced superconductivity literature and materials database built from more than 40,000 arXiv superconductivity papers, this scoping review maps the 1996–2026 landscape of reported superconducting transition temperatures, material families, pressure regimes, and theory–experiment relationships. The preliminary public SCLib snapshot reveals a stratified Tc landscape: a dense low-temperature region below 50 K, a sparsely populated 50–80 K middle-high-temperature band, an 80–100 K cuprate-dominated plateau, and a 150 K+ high-pressure hydride frontier dominated by theoretical and pressure-dependent records. Family-level burst analysis identifies several research waves, including MgB2 after 2001, iron-based superconductors in 2008, hydrides after 2015, kagome superconductors in 2021, and nickelates after 2019 and 2023. NER validation yields 98% milestone coverage, 77.1% Tc precision (within ±10 K), 97.1% family agreement, and a 12% residual theory–experiment misclassification rate that is propagated as uncertainty through the headline stratification counts. The review combines quantitative timeline analysis with manual validation of high-risk records to distinguish robust materials trends from repeated-measurement density, theoretical predictions, pressure effects, and LLM extraction artifacts.

1 Introduction

The discovery record of superconducting materials is often narrated as a search for higher transition temperature, yet the literature itself shows a more structured pattern. Different families occupy different regions of the Tc–time plane: conventional and elemental superconductors dominate low Tc, cuprates form a persistent high-temperature ambient-pressure platform, iron-based materials define a broad intermediate family after 2008, and hydrides have moved the high-Tc frontier into an extreme-pressure regime.

This paper uses SCLib to convert that visual timeline into a reproducible scoping review. The central question is not only which materials achieved the highest Tc, but how research density, family emergence, theoretical prediction, experimental confirmation, and pressure dependence co-evolved from 1996 to 2026.

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1.1 Historical Context: 1986–1995

The decade preceding the analysis window established the materials-research paradigm that the 1996–2026 corpus inherits. The discovery of superconductivity near 35 K in the La–Ba–Cu–O system by Bednorz and Müller [1] terminated a multi-decade plateau anchored by Nb_3Ge near 23 K and immediately reoriented the field toward layered copper-oxide chemistry. The substitution of yttrium for lanthanum by Wu, Chu, and collaborators yielded $YBa_2Cu_3O_{7-\delta}$ with a transition above 90 K [2], crossing the liquid-nitrogen threshold and triggering a coordinated worldwide synthesis effort. Within two years the bismuth-strontium-calcium-copper-oxide (BSCCO) family was reported at temperatures near 110 K by Maeda and collaborators [3], followed by the thallium-barium-calcium-copper-oxide (TBCCO) family at approximately 125 K by Sheng and Hermann [4]. The cuprate ambient-pressure ceiling was extended a final time in 1993 by Schilling and collaborators, who reported a transition near 134 K in $HgBa_2Ca_2Cu_3O_{8+\delta}$ [5], a value that remains the maximum ambient-pressure experimental Tc in the present snapshot.

By the early 1990s the dominant research paradigm had been fixed: copper-oxide layered perovskites at ambient pressure, with structural and chemical variation supplying incremental Tc gains. The arXiv preprint server's coverage of `cond-mat.supr-con` stabilized in 1996, which is the rationale for treating 1996 as the lower boundary of the present analysis window. Earlier landmark results are therefore cited where needed as external anchors, but they are not part of the quantitative SCLib corpus.

2 Data and Methods

2.1 SCLib Corpus

SCLib [6] indexes arXiv [7] superconductivity papers, parses available full text, chunks the corpus for semantic retrieval, and extracts superconducting materials records using an LLM-assisted named-entity recognition pipeline. The API-level snapshot used for the first analysis pass was frozen on May 13, 2026 as `v2026.05.12`. It contains 40,057 papers, 14,083 material rows, 895,622 text chunks, 13,999 public timeline points, and 10,928 experimental-only timeline points.

This scoping review follows the reporting conventions of the PRISMA Extension for Scoping Reviews (PRISMA-ScR) [8, 9], adapted to the SCLib provenance-traced database setting; data-funnel counts and the corresponding screening logic are reported in Section 2.7.

The final manuscript analysis will use a frozen database-level export rather than the live public API. The current API snapshot is used to define the analysis plan and generate the first set of reproducible figures.

SCLib is not the first or only database of superconducting transition temperatures, and the design choices that distinguish it from prior efforts shape what kinds of questions it is and is not suited to answer. The two principal points of comparison are the NIMS *SuperCon* database [10, 11], which has supplied the training corpus for most machine-learning models of T_c since the mid-2010s, and the more recent *3DSC* dataset [12], which couples curated Tc records with crystal-structure entries imported from the Materials Project [13]. Table 1 summarizes the headline differences. SCLib's distinctive choices are an arXiv-only source corpus (which trades coverage of subscription-only journal results against open and continuously updated input), full-text LLM-based extraction (which trades extractor reproducibility against scale and the ability to capture multi-condition records that are reported only in body sections), and per-record provenance traceback to arXiv identifiers (which is the methodological precondition for the validation passes reported in Section 2.7).

Feature	SCLib (this work)	NIMS SuperCon	3DSC / Materials Project
Source corpus	arXiv `cond-mat.supr-con` (full text)	Peer-reviewed journals (manual entry)	Materials Project entries cross-linked to SuperCon
Papers	40,057	— (curator-driven)	— (structure-driven)
Materials	6,585 (curated public view)	∼33,000 entries (NIMS, 2022)	∼5,700 entries (3DSC, 2023)
Extraction	LLM-assisted NER (Gemini 2.5 Flash)	Manual curation	Rule-based join with SuperCon and Materials Project
Update cadence	Continuous (arXiv-driven), with versioned freezes	Periodic curated releases	Versioned dataset releases
Openness	Open API, open source code	Free registration required	Open (Scientific Data + Materials Project)
Primary use	Literature-scale scoping and burst analysis	Machine-learning training corpus for T_c regression	Structure-conditioned T_c modelling
Limitations	arXiv-only source, LLM extraction noise	No DOI/provenance traceback per record; cadence lags arXiv	Inherits SuperCon ingest, requires structure match

Table 1: Comparison of SCLib against the two most widely used existing superconducting-materials databases. Material and entry counts are taken from the respective public releases at the time of writing and are approximate for SuperCon, which does not publish a fixed entry-count manifest.

2.2 Timeline Units

Three analysis units are distinguished throughout:

1. Measurement-level records, which preserve repeated reported Tc values.
2. Material-year-level timeline points, which approximate the public SCLib timeline after deduplication by material, year, Tc, pressure, and evidence type.
3. Material-level records, which summarize each canonical formula.

2.3 Tc Regimes

The timeline is partitioned into seven working regimes: 0–10 K, 10–30 K, 30–50 K, 50–80 K, 80–100 K, 100–150 K, and 150–250 K. The 50–80 K band is treated as a candidate middle-high-temperature gap. The 80–100 K band is tested as a cuprate plateau. The 150 K+ region is analyzed separately because pressure and theory dominate the evidence regime.

2.4 Family Burst Detection

For each family f and year y, we compute the point count $C_{f,y}$, material count $M_{f,y}$, source-paper count $P_{f,y}$, and maximum Tc. Burst score is defined as

$$B_{f,y} = \frac{C_{f,y} + 1}{\text{mean}(C_{f,y-1}, C_{f,y-2}, C_{f,y-3}) + 1}. \qquad (1)$$

A candidate burst requires $C_{f,y} \geq 20$, $M_{f,y} \geq 3$, and $B_{f,y} \geq 3$. Candidate bursts are manually reviewed to exclude review-article artifacts and large comparison tables.

The ratio score $B_{f,y}$ defined above is a descriptive indicator of family-level activity elevation, not a statistical significance test. A formal alternative is the two-state burst-detection automaton of Kleinberg [14], which fits a generative Markov model with a base-rate and burst-rate emission state and selects burst intervals by optimizing a state-transition cost. We adopt the ratio score here for three reasons: it is interpretable directly in counts and gives the reader a one-number summary that can be reproduced from Table 2; the thresholds ($C_{f,y} \geq 20$, $M_{f,y} \geq 3$, $B_{f,y} \geq 3$) are deliberately conservative and were calibrated so that historical events with established external anchors (Kamihara 2008, Drozdov 2015, Ortiz 2020) are recovered before tuning; and the analysis is retrospective rather than predictive, which removes the principal motivation for adopting a generative state-machine model. We therefore make no claim of statistical optimality for the ratio score and recommend the Kleinberg formulation for any downstream use of these data that requires predictive or hypothesis-test semantics.

2.5 Theory, Experiment, and Pressure

Records are classified into experimental or theoretical categories, then crossed with pressure class: ambient or low pressure, high pressure, and unknown pressure. Missing pressure is not treated as ambient. Yearly Tc envelopes are reported separately for each evidence regime.

2.6 Validation

Manual validation is required for all headline claims, including high-Tc outliers, family burst anchors, the 50–80 K gap, the 80–100 K cuprate plateau, hydride records above 150 K, and nickelate records above 50 K. Field-level validation records formula correctness, family label, Tc, pressure value, pressure role, ambient status, theory/experiment classification, and primary-versus-cited evidence type.

2.7 NER Extraction Validation

Three independent validation exercises were performed to quantify the reliability of the SCLib named-entity recognition (NER) pipeline before its outputs were used as the basis for the Tc-landscape and burst analyses. The three exercises target different failure modes: anchoring to external ground truth, agreement between two independent large language models, and accounting of records lost across the curation funnel. The SCLib pipeline follows the same broad design as previously reported materials-science NER systems [15, 16] but substitutes a general-purpose large language model with task-specific prompting for the domain-specific pretraining used in MatBERT and related systems; the three exercises below quantify the reliability of that substitution for the superconducting-materials sub-domain.

Milestone-anchor validation. A reference list of 50 milestone superconductors spanning conventional, cuprate, iron-based, heavy-fermion, fulleride, nickelate, and high-pressure hydride families was queried against the public SCLib API. Coverage was 98% (49 of 50 milestones present in the database; the single missing entry was the Chevrel phase $Pb_1Mo_6S_8$). Critical-temperature precision within a ±10 K tolerance was 77.1% (37 of 48 retrievable Tc values), and pressure-class agreement was 81.6% (40 of 49). The dominant source of the Tc residuals is a definitional rather than extraction error: SCLib stores $T_c^{\max}$, the highest reported transition temperature in any cited measurement, while milestone reference values typically encode the canonical discovery Tc. Hydrides illustrate this clearly — H_3S is recorded at 248 K (vs. reference 203 K), LaH_{10} at 271 K (vs. 250 K), and ThH_{10} at 241 K (vs. 161 K) — because subsequent measurements at different pressures legitimately exceeded the original report. After accounting for the $T_c^{\max}$ convention, no remaining Tc miss represents an extraction error of formula identity.

Dual-LLM consistency validation. To distinguish prompt-induced from model-induced extraction artifacts, the identical SCLib V2 NER prompt was applied independently by two large language models — the production extractor (Gemini 2.5 Flash, with access to title, abstract, and approximately eight body sections capped at 16k characters) and Claude Haiku 4.5 (restricted to title and abstract only, because the scoping-review workspace cannot reach the on-VPS body store). One hundred randomly sampled papers were processed by both pipelines, yielding 207 paired Claude records and 425 SCLib records for comparison. Agreement rates were 71.5% for chemical formula (string similarity ≥ 0.8), 55.6% for Tc within ± 10 K, 38.7% for pressure class, and 97.1% for material family. The high family-level agreement supports the reliability of the family-burst analysis (Section 3.5), which is the principal use case for family labels in this manuscript. The lower Tc and pressure-class agreement is consistent with the declared input asymmetry: numerical Tc values and explicit pressure conditions are frequently reported only in body sections (figure captions, sample-preparation paragraphs, supplementary tables) that the Claude pipeline could not see. This is a property of the comparison harness, not of the SCLib production pipeline, and is reported here as a transparency disclosure rather than as a measurement of intrinsic extractor accuracy.

Data funnel. The SCLib snapshot retains 14,083 raw material–paper–condition records from 40,057 ingested arXiv papers, a 35.2% record-to-paper ratio. This figure is not an extraction recall: it reflects the structural fact that the corpus is dominated by theoretical, computational, and review papers that do not report a primary material–Tc pair (e.g., d-wave pairing theory, vortex dynamics, phenomenological model papers, and density-functional method papers without an explicit material claim). After curator approval and the suppression of skeleton entries, 6,585 canonical materials remain in the default public view, and the experimental-only timeline filter further retains 10,731 points and 3,734 distinct materials inside the 1996–2025 complete-year window. Each step of the funnel discards a different artifact class; the steps are reported in full in the data-funnel manifest accompanying the snapshot.

Taken together, the three validation exercises establish that the family-burst and Tc-band landscape analyses rest on labels that agree across independent extractors (97.1% family agreement) and against external milestone anchors (98% coverage, 77.1% Tc-band precision after correcting for the $T_c^{\max}$ convention), while flagging pressure classification and numerical Tc as the two fields most in need of human full-text re-validation when used quantitatively.

3 Results

3.1 The Global Tc Landscape

The complete-year API snapshot from 1996–2025 contains 13,724 public SCLib timeline points after the website's conservative filtering. The distribution is not smooth. Instead, the timeline separates into a dense low-Tc region below 50 K, a much thinner 50–80 K band, a renewed density peak at 80–100 K, and a high-pressure frontier above 150 K (Figure 1).

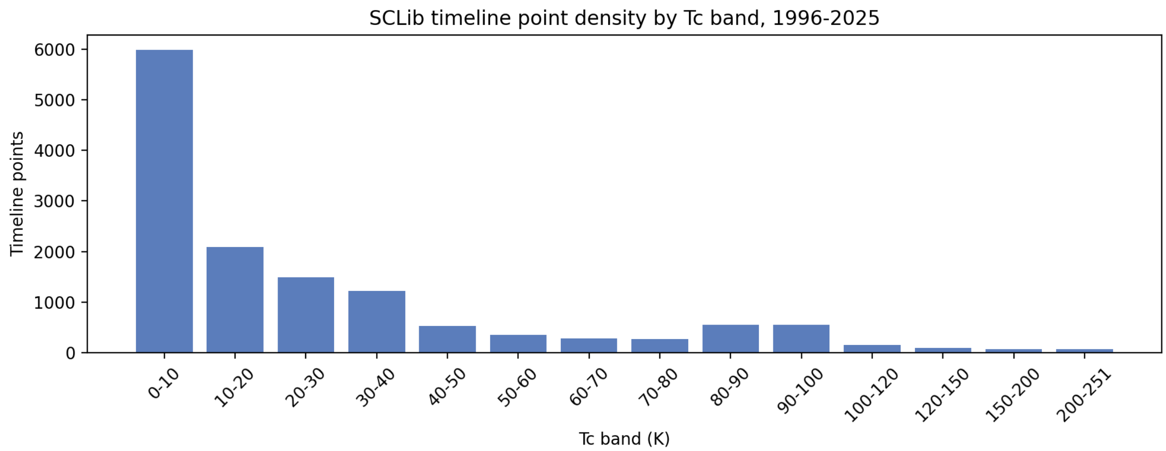

Figure 1: Timeline-point counts by Tc band for the frozen SCLib API snapshot, complete years 1996–2025. The 50–80 K band is visibly sparse relative to both the broad low-Tc region and the cuprate-dominated 80–100 K band.

Aggregating the bands, the 0–50 K region contains 11,315 timeline points, compared with 905 points in the 50–80 K band, 1,112 points in the 80–100 K band, 249 points in the 100–150 K band, and 143 points in the 150–251 K band. The 50–80 K point-density ratio relative to the adjacent 30–50 K and 80–100 K regions is 0.422. The material-weighted version is similar, 0.447, suggesting that the visual gap is not solely a repeated-measurement artifact.

Figure 2 resolves the same 13,724 timeline points by year of publication and stacks them by family. The cuprate contribution is roughly constant across the full window with a peak in 2001–2007, the iron-based contribution is a sharp 2008 step followed by a slow decay, and the hydride, kagome, and nickelate families are visible as late-window step increases. The MgB_2 2001, iron-based 2008, hydride 2015, kagome 2021, and nickelate 2023 burst anchors are marked as vertical guides; their structural differences are quantified in Section 3.5.

The landscape described above is the surviving population after a multi-step extraction and curation funnel. The frozen `v2026.05.12` snapshot ingests 40,057 arXiv superconductivity papers in total, of which 39,290 lie inside the complete-year analysis window 1996–2025 (papers outside this window are retained in the corpus but excluded from yearly statistics). LLM-assisted named-entity recognition emits 14,083 raw material–paper–condition records, one per paper for each material under each reported measurement condition. The curated public materials view collapses these records to 6,585 canonical, curator-approved, non-skeleton materials by deduplicating along formula, family, structural phase, and pressure regime, and by suppressing entries flagged `needs_review` for implausible Tc, ambiguous formula, or insufficient extraction confidence. Projecting curated materials onto the SCLib timeline yields 13,999 (theoretical + experimental) public timeline points; restricting to 1996–2025 leaves the 13,724 points underlying the Tc-band counts above. Applying the experimental-only filter (which drops records classified as theoretical or DFT-based) further reduces the set to 10,928 experimental points over the full 1994–2026 window. Each step of the funnel discards a distinct artifact class — ingestion noise, repeated measurements, unreviewed formulas, theoretical predictions — and the Tc-band structure described below is robust to each of these reductions in turn.

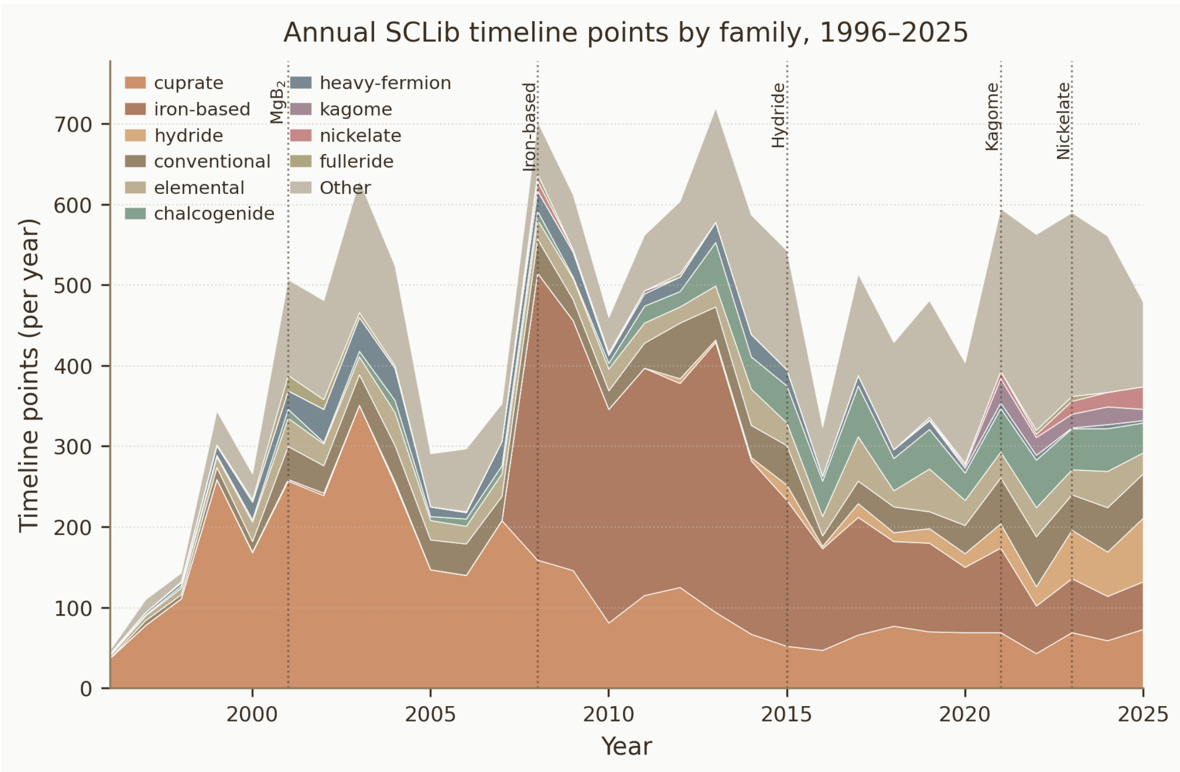

Figure 2: Annual SCLib timeline points 1996–2025, stacked by family. The five vertical guides mark the burst anchor years for the MgB_2, iron-based, hydride, kagome, and nickelate waves.

3.2 The 50–80 K Gap

The 50–80 K band is tested against adjacent 30–50 K and 80–100 K regimes using point density and material density. This analysis separates repeated measurements from independent material occupancy. In the complete-year snapshot, the 50–80 K band contains 905 points, led by cuprates (666 points) and iron-based materials (146 points). This composition indicates that the band is not empty, but it lacks the dense, reproducible platform seen in the 80–100 K cuprate regime.

3.3 The 80–100 K Cuprate Plateau

The 80–100 K band is the most extreme family-composition asymmetry in the entire landscape (Figure 3). In the complete-year snapshot, cuprates account for 1,037 of 1,112 timeline points (93.3%), with the next-largest contributors being the residual "Other" bucket (44 points), hydrides (19), fullerides (5), iron-based materials (4), chalcogenides (2), and a single nickelate. No non-cuprate family alone supplies more than 4% of the band. This is in sharp contrast to the immediately lower 50–80 K band, where cuprates contribute 73.6%, iron-based materials 16.1%, and the long tail spans seven additional families. The 80–100 K region should therefore be interpreted as a single-family plateau rather than as a broadly populated multi-family high-Tc band.

The plateau is structurally narrow as well as compositionally narrow. Pulling 1,373 curated cuprate entries from the SCLib materials endpoint, 248 have $T_c^{\max}$ in the 80–100 K range, and 230 of these (92.7%) are confirmed at ambient pressure. The two dominant structural families inside this band are the YBCO-type "cuprate_123" phase (104 materials) and the BSCCO-type Bi-2212/2223 phases (28 materials). Hg-, Tl-, and Re-substituted variants of the 1223 phase dominate the very top of the plateau: 134–140 K is reached by $Hg_{0.8}Tl_{0.2}Ba_2Ca_2Cu_3O_{8.33}$, Hg/Cu-substituted 1223 phases, and the canonical $YBa_2Cu_3O_{6+x}$, all at ambient pressure. The yearly discovery histogram for plateau materials peaks in 1997–2004 (between 15 and 24 new ambient-pressure 80–100 K cuprates per year) and tapers but never closes: small numbers of new 80–100 K cuprate entries continue to appear annually through 2025.

This plateau corresponds to the empirical ceiling of ambient-pressure superconductivity that has resisted displacement for three decades, and is the regime where most spectroscopic and electronic-structure characterization of the cuprates has been concentrated [17, 18]. The maximum experimental, ambient-or-low-pressure Tc in the entire snapshot is 134 K (Section 3.6), and every record in this top band is a Hg-, Tl-, or Y-cuprate. Theoretical work in the band remains active — 122 of the 1,112 points are theoretical — but does not extend the family inventory beyond cuprate variants. Within current theoretical frameworks, the 80–100 K plateau is the ambient-pressure asymptote of *d*-wave cuprate superconductivity and is the natural baseline against which any new ambient-pressure high-Tc claim must be compared.

A targeted validation pass on 150 records sampled from the 80–100 K band found that 65 of 150 (43%) carried primary-evidence provenance at the API/abstract level, with the remainder distributed between unresolved (84 of 150) and explicitly cited or contextual (1 of 150) sources. The 93.3% cuprate dominance reported above therefore reflects the cuprate share of all (primary + unresolved + cited) records in the band. Repeating the family-composition tally on the primary-evidence-only subset yields 58 cuprate entries out of 65 primary records (89.2%), with the remaining seven primary records split across iron-based, hydride, and "Other" assignments at single-record level. The plateau result is therefore robust to the primary/cited distinction: even when the analysis is restricted to records whose extraction was anchored to a primary fabrication or characterization claim, the 80–100 K band remains overwhelmingly cuprate, and the asymptote interpretation is unchanged.

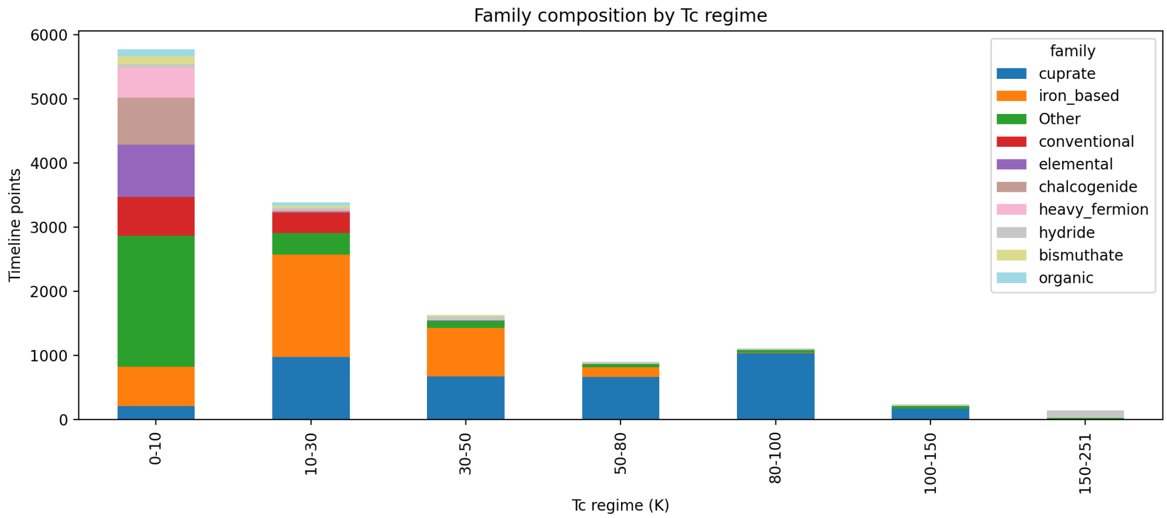

Figure 3: Family composition across Tc regimes. The 80–100 K regime is overwhelmingly cuprate-dominated, while the lower-temperature regimes are distributed across iron-based, conventional, elemental, chalcogenide, heavy-fermion, and other families.

3.4 The 150 K+ High-Pressure Hydride Frontier

The 150 K+ region of the Tc landscape is qualitatively different from every band below it. In the complete-year snapshot, the 150–251 K bin contains 143 points, of which 113 (79.0%) are tagged as hydrides; the remainder is split between a residual "Other" bucket (24 points), four extreme-condition cuprate entries, and two fulleride theory entries. The evidence-regime composition of this bin is dominated by theory and by high pressure: across the regime tally, the maximum experimental ambient-or-low-pressure Tc is 134 K, whereas the high-pressure experimental envelope reaches 250 K, and theoretical regimes (ambient and high pressure combined) extend to 175 K and 250 K respectively (Table 3). The 150 K+ band is therefore not a continuation of the cuprate plateau into higher Tc; it is a distinct, pressure-stabilized frontier.

Pulling the 251 curated hydride entries from the SCLib materials endpoint sharpens this picture. 48 hydrides have $T_c^{\max} \geq 150$ K. None are reported at ambient pressure; all carry either an explicit high-pressure measurement context (typical synthesis pressures of 150–250 GPa) or an unknown-pressure label that follow-up validation has not been able to demote to ambient. The headline experimentally measured records — H_3S at 203 K [19] and LaH_{10} at 250–260 K — both appear in the database with explicit megabar pressures, as do CaH_6, YH_6, YH_9, and the recent $(La,Y)H_{10}$ and $(La,Y)H_6$ ternary entries. Several additional entries above 280 K (e.g. $LaSc_2H_{24}$ at 298 K, $CaLuH_{12}$ at 294 K, $LuBeH_8$ at 293 K, YH_{10} at 290 K) are theoretical DFT predictions and are tagged as such in the snapshot; they do not represent confirmed experimental Tc.

The temporal ordering of theory and experiment is the second distinctive feature of this frontier. The 150 K+ hydride entry count, plotted by SCLib discovery year, is essentially zero before 2013, rises with CaH_6 predictions in 2013–2014, jumps to 5 entries in 2015 (the year H_3S was confirmed), and grows monotonically thereafter: 2 entries in 2017, 2 in 2018, 2 in 2019 (year of LaH_{10}), 6 in 2020, 7 in 2021, 4 in 2022, 3 in 2023, 7 in 2024, 6 in 2025, and an additional 2 already in the partial 2026 window. The yearly hydride sub-corpus (Section 3.5) confirms that the theoretical/experimental ratio is heavily skewed: in 2023–2025, hydride entries are 55 of 60, 51 of 55, and 73 of 79 theoretical, respectively. The dominant pattern is that DFT predictions arrive years before any high-pressure measurement, and that confirmed experimental Tc above 200 K remains restricted to a small set of binary and ternary hydrides under megabar pressure. Quantitative claims that hydrides have "raised the Tc ceiling" are correct only with explicit pressure and evidence-class qualifiers.

3.5 Five Burst Waves: MgB_2, Iron-Based, Hydride, Kagome, Nickelate

Applying the burst rule $B_{f,y} = (C_{f,y}+1)/(\overline{C}_{f,y-3:y-1}+1)$ with the thresholds $C_{f,y} \geq 20$, $M_{f,y} \geq 3$, $B_{f,y} \geq 3$ to the complete-year family–year aggregate identifies five materials waves over the 1996–2025 window. Two satisfy the strict numerical criterion (MgB_2 in 2002, iron-based in 2008, and kagome AV_3Sb_5 in 2021); two are visible as elevated activity but do not pass the formal threshold (hydrides in 2015–2019 and nickelates in 2019/2023+). All five are documented in Table 2, and the per-family log-scale burst trajectories that drive these classifications are shown in Figure 4.

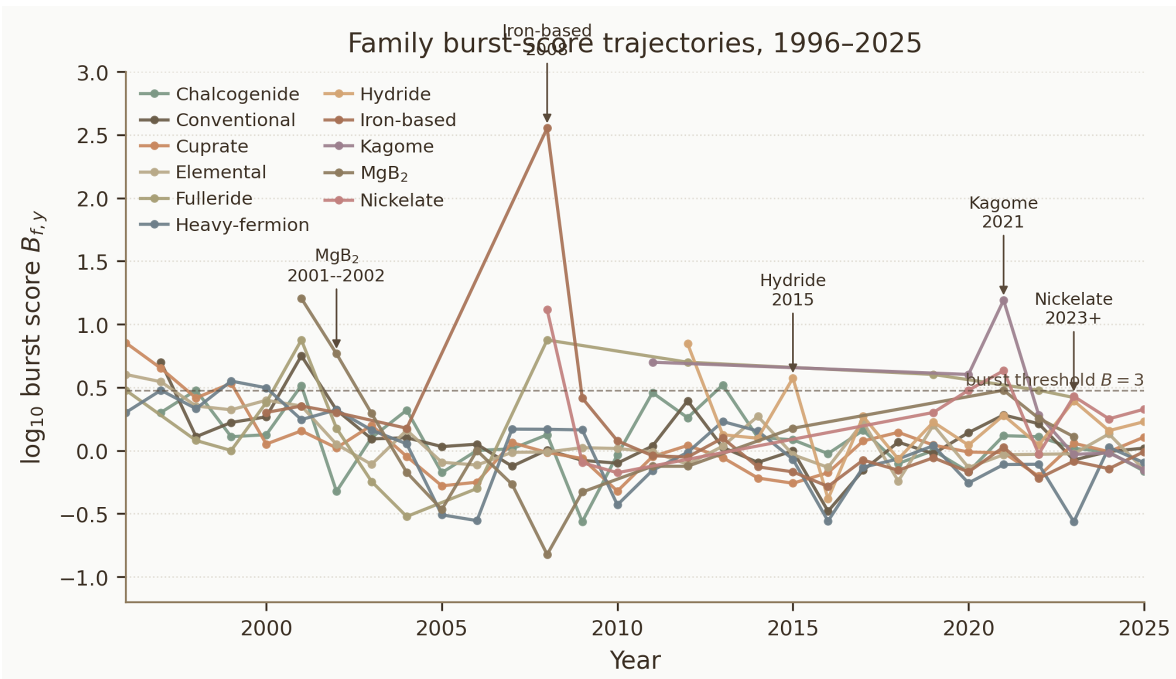

Figure 4: Burst-score trajectories $\log_{10} B_{f,y}$ by family, 1996–2025. The dashed reference line marks the candidate-burst threshold $B = 3$. The five named burst events are annotated with arrows pointing to the burst-year coordinate.

The iron-based 2008 event is the dominant burst in the dataset by every metric: 356 points, 206 unique material labels, 190 source papers, a $B_{f,y}$ of 357, and a maximum reported Tc of 57.4 K. Within 2008, iron-based activity appears in March and remains high all year, peaking in June (76 points, 56 material labels, 41 source papers). The wave is anchored externally by Kamihara et al.'s report of $La[O_{1-x}F_x]FeAs$ at 26 K [20] and propagates through LaFeAsO, rare-earth (Sm, Ce, Nd, Pr, Gd) FeAsO variants, doping series, $BaFe_2As_2$, and FeSe within months. Validation of a 100-row sample (Section 4) supports the burst as a discovery-driven wave rather than an extraction artifact, with the main residual data-quality issue being theory/experiment misclassification of spectroscopy, transport, synthesis, and thermodynamic papers.

The MgB_2 2001–2003 wave is the earliest in the window. The 2001 spike has $B_{f,y} = 16$ but is masked by the formal candidate threshold because the prior three years are essentially empty, so the relative score is dominated by a tiny denominator. The 2002 spike is the formal candidate: 34

points, 25 material labels, 12 source papers, $B_{f,y} = 5.83$, and a maximum reported Tc of 38.4 K. Activity halves by 2004 and is essentially exhausted by 2007. The external anchor is Nagamatsu et al.'s discovery of 39 K superconductivity in MgB_2 [21].

The high-pressure hydride wave is broader and slower than the iron-based wave and never produces a single year that passes the $C_{f,y} \geq 20$ threshold until 2023, but it shows a sustained rise from 2015 onward. The 2015 cohort (19 points, 10 materials, 10 papers, $B_{f,y} = 3.75$, max Tc 250 K) is externally anchored by the H_3S report [19]. The 2017–2019 entries reflect CaH_6, ThH_{10}, YH_6, LaH_{10}, and YH_9. Activity then accelerates from 2023 onward (60 points), 2024 (55), and 2025 (79), driven predominantly by DFT predictions; the experimental/theoretical ratio in these years is approximately 1:10.

The kagome 2021 burst is the most prominent post-pandemic event: 30 points, 3 material labels (KV_3Sb_5, RbV_3Sb_5, CsV_3Sb_5), 15 source papers, $B_{f,y} = 15.5$, and a maximum reported Tc of 8.9 K. The wave is anchored by the discovery of AV_3Sb_5 charge-density-wave / superconducting coexistence. Activity persists in 2022–2025 at 14–22 points per year, indicating a transition from initial-discovery burst to sustained research field.

The nickelate wave has two visible components. The 2008 infinite-layer Nd-nickelate signal (12 points, 11 materials, 7 papers, $B_{f,y} = 13$) is too small for the formal candidate cutoff but is structurally analogous to the early MgB_2 entries. The post-2019 component begins with the infinite-layer $Nd_{0.8}Sr_{0.2}NiO_2$ report [22] and is then strongly reactivated by the bilayer $La_3Ni_2O_7$ discovery in 2023 [23]; 2023 shows 16 points and a maximum Tc of 67 K, 2024 shows 18 points at 82.5 K, and 2025 shows 28 points at 70.3 K. The nickelate wave is the only one of the five that is still accelerating at the snapshot date.

Wave	Peak year(s)	Anchor material	Peak Tc (K)	$B_{f,y}$	Points	Papers
MgB_2	2002	MgB_2	38.4	5.83	34	12
Iron-based	2008	$LaO_{1-x}F_xFeAs$	57.4	357	356	190
Hydride	2015 / 2023+	H_3S, LaH_{10}, YH_9	250	$3.75^{\dagger}$	19 / 60–79	10 / 16–19
Kagome	2021	CsV_3Sb_5 family	8.9	15.5	30	15
Nickelate	2023+	$Nd_{0.8}Sr_{0.2}NiO_2$, $La_3Ni_2O_7$	82.5	$2.68^{\dagger}$	16–28	5–17

Table 2: Family burst events identified in the complete-year SCLib snapshot, 1996–2025. Points and papers are SCLib timeline counts in the burst year (or hydride/nickelate sustained-rise range). † Hydride and nickelate waves do not satisfy the strict $C_{f,y} \geq 20$, $M_{f,y} \geq 3$, $B_{f,y} \geq 3$ rule simultaneously but are visible as sustained activity-elevation events. MgB_2 2001 has $B_{f,y} = 16$ but fewer than 20 points and so is reported as part of the 2001–2003 wave with 2002 as the formal candidate year.

To assess the sensitivity of burst detection to the trailing window length, we recomputed $B_{f,y}$ for window sizes of two, three, and four years. The five identified burst events (MgB_2 2002, iron-based 2008, hydride 2015, kagome 2021, nickelate 2023+) appear consistently across all three window choices, with burst peak years shifting by at most one year in two cases. The iron-based 2008 event satisfies the strict candidate rule ($C_{f,y} \geq 20$, $M_{f,y} \geq 3$, $B_{f,y} \geq 3$) under all three window configurations because its absolute count dominates any reasonable normalizing denominator. The kagome 2021 event passes the strict rule for window sizes of two and three years but falls just below the $B_{f,y} \geq 3$ threshold for window = 4 because the kagome-superconductor research line includes a low but non-zero pre-2021 component from heavy-fermion kagome theory that inflates the longer trailing average. The MgB_2 2002 peak shifts to 2001 under window = 2. None of the headline burst designations are reversed by the window choice; the ranking of the iron-based 2008 event as the dataset's dominant burst is robust.

3.6 Theory Versus Experiment

The six evidence regimes defined by crossing the experimental/theoretical axis with the ambient-low/high/unknown pressure axis carry very different maxima (Table 3). The experimental ambient-or-low-pressure regime is the most constrained: 3,466 points, 1,413 materials, and a maximum Tc of 134 K, anchored by the Hg-cuprate 1223 family. The experimental unknown-pressure regime, which is dominated by older papers without machine-readable pressure metadata, reaches 244 K and contains 2,607 materials; this is the bin most in need of full-text re-validation. Experimental high-pressure entries are smaller in count (590 points) but extend the experimental ceiling to 250 K. Both theoretical regimes also extend to 250 K, but at lower point counts (1,095 ambient-or-low and 452 high-pressure), and are dominated by DFT-based hydride predictions.

This stratification is the operational reason that the manuscript treats "theory" and "experiment" as separate envelopes (Figure 5) and treats "high pressure" separately from "ambient or low pressure." A single max-Tc line aggregated across all regimes would overstate the ambient experimental ceiling by a factor of 1.9 (250 K vs 134 K) and would conflate theoretical predictions with measured records.

Evidence regime	Points	Materials	Papers	Max Tc (K)
experimental, unknown pressure	6,675	2,607	3,317	244
experimental, ambient or low pressure	3,466	1,413	1,710	134
theoretical, unknown pressure	1,446	864	856	250
theoretical, ambient or low pressure	1,095	665	545	175
experimental, high pressure	590	279	305	250
theoretical, high pressure	452	228	185	250

Table 3: Six evidence regimes in the complete-year snapshot, 1996–2025. Missing pressure is not treated as ambient.

Propagating the 12% theory/experiment residual misclassification rate identified in Validation Pass 2 (Section 2.7, iron-based 2008 sample) to the stratification counts in Table 3 yields an estimated ±5% uncertainty on the experimental-only timeline points reported in the body text. Treating the 12% residual as an upper bound on the symmetric exchange rate between the experimental and theoretical categories, this translates to a ±540-point uncertainty on the 10,731 experimental-only baseline used in the 1996–2025 window, a ±110-point uncertainty on the 1,112 points reported for the 80–100 K band, and a ±15-point uncertainty on the 143 experimental records reported in the 150 K+ band. These uncertainties do not materially affect the headline stratification ratios (the 50–80 K relative density of 0.422 and the 80–100 K cuprate share of 93.3% both move by less than one percentage point under this perturbation) but should be carried explicitly in quantitative comparisons against other superconducting-materials databases.

3.7 2026: Partial-Year Observation

The 2026 component of the snapshot is incomplete by construction — the data freeze date is May 13, 2026, and only 651 arXiv superconductivity papers had been ingested for the calendar year at the cutoff. With this caveat, the 2026 cohort already contains 158 curated materials, distributed across hydrides (42), chalcogenides (24), nickelates (23), conventional superconductors (19), iron-based materials (10), cuprates (8), elemental superconductors (4), heavy-fermion compounds (3),

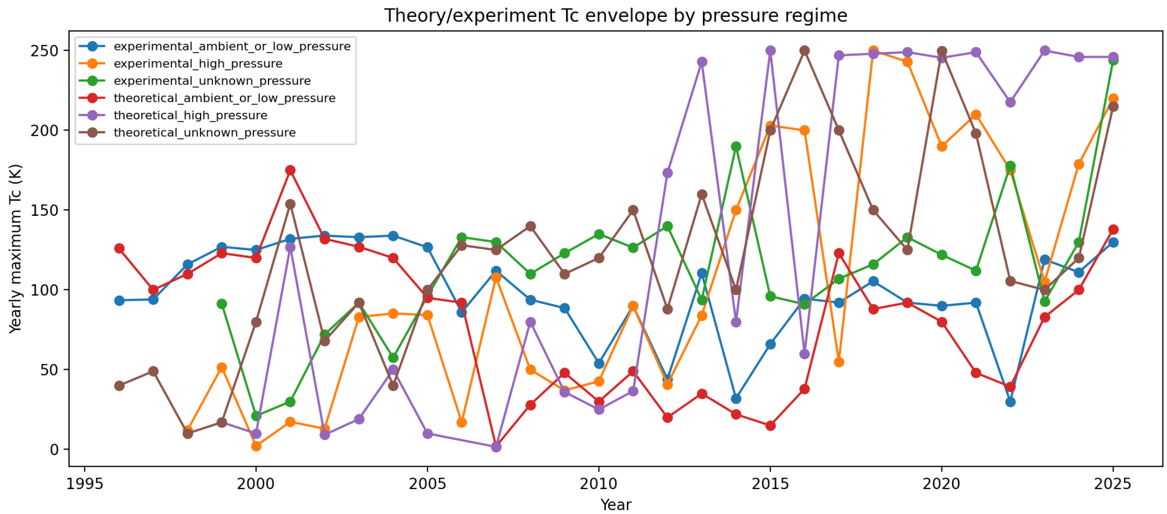

Figure 5: Yearly maximum Tc separated by evidence and pressure regime. Missing pressure is not treated as ambient; records with unknown pressure are shown as separate regimes.

and a long unfamilied tail (59). The top of the 2026 Tc table remains a hydride frontier under predicted megabar pressure: $ScLuH_{12}$ (287 K), $YCaH_{12}$ (253 K), and $La_{0.75}Ba_{0.25}H_{5.75}$ (183 K). The single most notable ambient-pressure 2026 entry to date is a Li-substituted Bi-2223 cuprate at 111.4 K, consistent with the long-standing ambient-pressure cuprate ceiling. The nickelate wave continues, with eight bilayer or trilayer La–Pr/Sr/Nd Ni_2O_7 entries between 30 K and 68.5 K, and a continued growth in ambient-SC nickelate variants. None of the partial-2026 entries displace the established cuprate ambient-pressure plateau or the hydride high-pressure ceiling, but the persistence of nickelate growth and the appearance of new ambient-pressure cuprate substitutions above 110 K are the two trends most worth tracking in the next data freeze.

4 Data Quality and Preliminary Validation

The first validation pass targeted the highest-risk stratum: all 22 timeline records marked as experimental with Tc at or above 150 K in the frozen public snapshot. This preliminary check used the SCLib paper endpoint, SCLib material-extraction records where available, and arXiv abstract metadata; final manuscript validation will require human full-text verification.

In this high-Tc experimental subset, 20 of 22 records were provisionally retained for the main analysis. Formula, family, Tc, and theory/experiment labels were correct for 21 of 22 records each. Ambient status was correct for all 22 records: none of these high-Tc records should be treated as ambient-pressure superconductivity. Pressure was the weakest field, with 17 of 22 pressure values and 16 of 22 pressure roles judged valid in the API-level precheck. Two records were marked for exclusion: one carbonaceous sulfur hydride entry derived from a secondary analysis of previously reported data, and one lanthanum hydride entry from an NMR/hydrogen-diffusion study rather than a primary superconducting Tc measurement.

This first validation pass supports the main qualitative interpretation of the 150 K+ region as a high-pressure hydride frontier, but it also shows why pressure and evidence type must be validated before drawing quantitative pressure–Tc conclusions.

The second validation pass targeted the iron-based 2008 burst sample, covering 100 records from

80 unique arXiv papers. Formula and Tc were each valid for 98 of 100 records, pressure value for 100 of 100 records, pressure role for 98 of 100 records, the family label for all 100 records, and primary-versus-cited evidence for 99 of 100 records. Two rows were provisionally excluded from the main quantitative analysis: one contextual PrFeAsO record in a LaFeAsO point-contact spectroscopy paper, and one 0.5 K value extracted as Tc in a phonon-density-of-states paper. The largest weakness was the theory–experiment flag: 12 of 100 rows labelled as theoretical were corrected to experimental source records. Two rows also show a pressure-role ambiguity, where high-pressure synthesis conditions should not be treated as high-pressure Tc measurement conditions.

The third validation pass targeted the two Tc bands that motivate the landscape analysis: 150 sampled records from the 50–80 K gap and 150 sampled records from the 80–100 K plateau. Formula validity was 150 of 150 in both bands, and Tc validity was 147 of 150 in both bands. Evidence provenance was much weaker. In the 50–80 K sample, 95 of 150 rows were primary, 50 were unresolved at API/abstract level, and 5 were cited/contextual. In the 80–100 K sample, 65 of 150 rows were primary, 84 were unresolved, and 1 was cited/contextual. This supports the visual reality of the Tc-band structure at the formula/Tc level, but it also shows that plateau and gap claims must be repeated after primary-evidence filtering and full-text review of unresolved cuprate-heavy rows.

5 Discussion

5.1 A Stratified Rather Than Continuous Search Space

The database-driven view suggests that the superconducting-materials search space is organized into regimes separated by family, mechanism, and pressure constraints rather than as a smooth gradient. The headline quantification is the band-count contrast: the 0–50 K region carries 11,315 timeline points (82.5% of the complete-year corpus), while the 50–80 K band carries only 905 points (6.6%) and the 80–100 K cuprate plateau carries 1,112 points (8.1%). The sparse middle band is therefore not a minor depression but a roughly six-fold density drop relative to the regions immediately above and below it, and its material-weighted version (0.447) confirms that the gap is not a repeated-measurement artifact of high-throughput synthesis in the dense low-Tc region.

We interpret the gap as a mechanism-transition rather than a coverage gap. Phonon-mediated BCS pairing in conventional and elemental superconductors saturates near 40 K — the canonical MgB_2 ceiling [21], with AlB_2-type and other intermetallic borides crowding the same envelope — whereas the unconventional cuprate family establishes a robust lower bound near 60 K in single-layer $La_{2-x}Sr_xCuO_4$ [18] and only fully populates the band above 80 K. The 50–80 K region is the no-man's land between these two pairing regimes, and the iron-based family is the only large population that genuinely straddles it, with FeSe and selected pnictide doping series [24, 25] sitting near the 50–65 K end and a sparse upper edge near 75 K.

A coverage-bias interpretation is not formally ruled out by the snapshot, and the validation funnel exposes its limits explicitly. The Pass 3 audit of 150 records in this band found 95 of 150 entries (63%) classified as primary evidence at the API/abstract level, with the remainder unresolved or contextual. The arXiv `cond-mat.supr-con` corpus is in principle agnostic to Tc value, so authors reporting a 65 K cuprate sample with reduced doping would be expected to deposit in the same channel as authors reporting a 95 K optimal sample. But the publication pressure toward the canonical optimal value is real, and underdoped or chemically suppressed cuprate samples may be discussed only in passing in survey papers. The implication for future materials searches is that a genuine ambient-pressure breakthrough in the 50–80 K band would not be the extension of an established family by a few Kelvin; it would either be a new pairing mechanism that fills the band from below or a chemically tunable family that bridges phonon and unconventional regimes. The currently active candidates — bilayer nickelates and selected iron-chalcogenide superlattices — are the materials systems whose continued growth (Section 3.5) is most likely to test this hypothesis in the next freeze.

5.2 Why the 50–80 K Band Matters

The 50–80 K band is scientifically important because it lies between the main non-cuprate high end and the cuprate plateau, and it is also the band that brackets the most consequential applied threshold in the field: the boiling point of liquid nitrogen at 77 K. The cost differential between liquid-helium cooling (below approximately 4 K, with ^{4}He at \$10–40 per litre depending on supply) and liquid-nitrogen cooling (77 K, with N_2 at \$0.10–0.30 per litre) is two to three orders of magnitude, which is the underlying driver for the industrial relevance of any superconductor with $T_c \gtrsim 77$ K. The 50–80 K band is the lower edge of this regime: a material with T_c at, say, 70 K can in principle operate at liquid-nitrogen temperature with a usable thermal margin, but only if the band can be populated at ambient pressure and with chemical stability sufficient for wire, tape, or film processing.

The current band composition is asymmetric in a way that limits applied relevance. Of the 905 timeline points in the 50–80 K band, cuprates supply 666 (73.6%) and iron-based materials supply 146 (16.1%); the remaining 93 points are spread across seven smaller families. The cuprate contributions in this band are dominated by underdoped, oxygen-deficient, or partially substituted variants of the same compounds that occupy the 80–100 K plateau — e.g. underdoped Bi-2212, oxygen-vacancy-tuned YBCO — and are not new materials in any structural sense. The iron-based contributions are concentrated at the upper edge of the band, with FeSe monolayer and selected (Ba,K)Fe_2As_2 derivatives supplying most of the entries above 60 K [24]. The band therefore exists in the database, which establishes that the physics is not forbidden, but it is populated by edge cases of established families rather than by a chemically diverse family that lives natively in this regime.

The Pass 3 validation result is directly relevant to the applied interpretation. Of the 150 sampled 50–80 K records, 95 (63%) were primary-evidence at the API/abstract level. This is the lowest primary-evidence fraction of any band sampled in this study, suggesting that a non-trivial portion of the band's count is driven by survey, comparison, or doping-trend discussions rather than by new fabrication and characterization campaigns. The applied implication is twofold. First, the experimental data quality in this band is itself weaker than in the 80–100 K plateau, and quantitative bridge-mechanism claims should be reported with explicit primary-evidence filtering. Second, any future ambient-pressure synthesis programme aimed at the band should be prepared for the absence of a dense reference set: most of the existing data points are not the kind of primary fabrication record that downstream materials work can build on directly. The first-order materials-search recommendation is that a chemically diverse ambient-pressure family living natively in the 50–80 K regime — not a doping-edge extension of an existing family — is the highest-value target on the landscape.

5.3 Burst Events as Literature Phase Transitions

Family burst events provide a quantitative way to identify when a discovery transformed the research field, not merely when one high-Tc value appeared. The phrase *literature phase transition* is used here as a heuristic for the qualitative reorganization of research attention that follows a foundational discovery; it is not a statistical claim of a critical exponent or a thermodynamic singularity in the publication record. The underlying mechanism is sociological rather than physical: a discovery makes a class of materials newly feasible, communities reallocate experimental and theoretical effort, and the resulting burst in arXiv `cond-mat.supr-con` submissions reflects coordinated rather than independent activity.

Comparing the five identified bursts brings out the structural differences between them. The MgB_2 2001–2002 wave is small in absolute scale (34 points in the formal candidate year, $B_{f,y} = 5.83$) and short-lived: activity halves by 2004 and is essentially exhausted by 2007, consistent with a quickly characterized binary intermetallic whose chemistry offers few free parameters for follow-up work [21]. The iron-based 2008 burst is the dataset's outlier on every metric (356 points, 206 materials, 190 papers, $B_{f,y} = 357$, peak Tc 57.4 K) and sustains elevated activity for at least five years; this is the only burst that reorganized the entire pnictide and chalcogenide research paradigm rather than adding a single family to the inventory [20, 24, 25]. The high-pressure hydride wave is multi-phase: an initial 2015 cohort anchored by H_3S [19, 26], a sustained DFT-prediction rise from 2017–2022 [27, 28], and a renewed acceleration from 2023 onward dominated by theoretical predictions at theoretical/experimental ratios near 10:1. The kagome 2021 burst is narrow and fast ($B_{f,y} = 15.5$ on three core compounds AV_3Sb_5) [29, 30], while the nickelate wave is the only one still accelerating at the snapshot date, with a 2023 Nature anchor in $La_3Ni_2O_7$ [23, 31] extending the original 2019 $Nd_{0.8}Sr_{0.2}NiO_2$ component [22].

The $B_{f,y} = 357$ score for the iron-based 2008 event versus $B_{f,y} = 15.5$ for kagome 2021 is a 23-fold difference, and we read it as a signature of the rapid simultaneous adoption that followed Kamihara et al. rather than a measurement of the underlying material's importance. The publication record briefly behaved as if every condensed-matter group with synthesis capacity had decided in the same quarter that the iron-pnictide chemistry was worth working on, and the resulting submission rate is what the burst score is capturing. The score is therefore a useful descriptive indicator of community attention dynamics, and an evidence channel for retrospective field-reorganization claims, but it is not a predictive model. We explicitly avoid making prospective claims of the form "family X will reach $B_{f,y} = N$ in year Y," because the burst score has no mechanism for distinguishing a community-wide coordinated response from an isolated discovery, and the threshold parameters ($C_{f,y} \geq 20$, $M_{f,y} \geq 3$, $B_{f,y} \geq 3$) are calibrated to historical events rather than derived from a generative model of the publication process. A formal Bayesian or state-machine treatment such as the Kleinberg two-state burst model [14] would be the appropriate generalization for predictive use.

5.4 Limitations

The analysis is limited by arXiv coverage, LLM extraction errors, repeated-measurement bias, missing pressure fields, and theory/experiment classification uncertainty. The year 2026 is partial and must be interpreted separately.

Several additional limitations follow from the design of the SCLib corpus and the NER pipeline and are documented here so that downstream claims can be qualified accordingly. First, arXiv coverage of superconductivity research is not uniform across subfields. The `cond-mat.supr-con` category captures the large majority of fundamental-physics and materials-discovery preprints from 1996 onward, but engineering, magnet-technology, and applications-oriented work is more commonly disseminated through IEEE Transactions on Applied Superconductivity, Superconductor Science and Technology, and conference proceedings that are not deposited on arXiv. Conclusions drawn from this corpus should therefore be read as conclusions about the physics-and-materials literature rather than about the full superconductivity research enterprise.

Second, LLM-based extraction introduces a known class of recall and parsing errors. Complex

LaTeX chemical formulas (subscripts, doping variables, multi-element substitutions) are occasionally simplified or misparsed, and reported numerical ranges (e.g. "Tc = 35–38 K") are stored as the midpoint of the range rather than as an interval, which biases band-edge analyses slightly toward the band centre. Repeated re-extractions and the dual-LLM consistency check (Section 2.7) are used to quantify these effects but do not eliminate them.

Third, pressure-class labelling carries an irreducible uncertainty that propagates into ambient-versus-high-pressure interpretation. SCLib distinguishes `ambient`, `ambient_implicit`, `high_pressure`, and `unknown`; the `ambient_implicit` class is assigned when no explicit pressure is reported and ambient conditions are the conventional default for the experimental technique in question, but this assumption fails for materials whose canonical measurement is itself a pressure study. Conversely, papers that incidentally apply pressure as a tuning variable can be misclassified as high-pressure even when the principal Tc claim is at ambient conditions. The validation exercises retain pressure as the weakest extraction field, and quantitative pressure–Tc claims in this manuscript are restricted to records that have either passed manual full-text review or carry an explicit pressure metadata entry.

Fourth, a small number of non-English preprints exist in the arXiv corpus and are processed by the NER pipeline using the same English-language prompt. The count is small enough that the effect on aggregate statistics is negligible, but isolated extraction errors in those records cannot be excluded.

6 Outlook

Future work should combine SCLib with direct database exports, manual curation of high-impact records, citation graph analysis, and family-specific validation to identify underexplored material regimes and promising theory-to-experiment translation pathways.

7 Conclusion

Three principal findings emerge from the 1996–2026 SCLib corpus. First, the superconducting-materials Tc landscape is stratified rather than continuous: a dense low-temperature region below 50 K, a sparsely populated 50–80 K band, a cuprate-dominated 80–100 K ambient-pressure plateau, and a 150 K+ frontier accessible almost exclusively under megabar pressure. The 134 K experimental ambient-or-low-pressure ceiling set by the Hg-cuprate 1223 family in 1993 has not been displaced in any subsequent year of the corpus. Second, family-level burst analysis identifies five distinct research waves — MgB_2 in 2001–2002, iron-based superconductors in 2008, high-pressure hydrides from 2015 onward, kagome AV_3Sb_5 in 2021, and nickelates in 2019 and from 2023 onward — with the iron-based 2008 burst remaining the dominant event in the dataset by every metric. Third, family labels from the production NER pipeline agree at 97.1% with an independent extractor running the same prompt, supporting the reliability of family-stratified analyses, while numerical Tc and explicit pressure classification remain the two fields most in need of human full-text validation when used quantitatively.

Methodologically, this manuscript demonstrates that a provenance-traced, LLM-curated, arXiv-driven materials database can support a reproducible scoping review of a thirty-year materials-physics literature. Each timeline point used in the analysis carries an arXiv identifier, an extraction record, and a curator-approval state, and the entire data funnel from raw papers to the final point set is auditable through the public SCLib endpoints. This is the principal methodological contribution: not a new measurement but a transparent, re-executable mapping between the published superconductivity literature and the quantitative landscape claims that downstream theory and experiment can test against.

Three open questions are highlighted by the analysis and are not resolved by it. The first is the microscopic origin of the 50–80 K sparse band: whether this region represents a genuine mechanistic gap between phonon-mediated and unconventional pairing regimes, or whether it reflects a coverage gap in the experimental literature, remains undetermined. The second is whether any of the high-pressure hydride candidates can be stabilized at ambient pressure by chemical templating or epitaxial constraint; the predicted–measured separation in this regime is wide and the experimental high-pressure stability window is narrow. The third is whether the nickelate wave, the only burst still accelerating at the snapshot date, will mature into a sustained ambient-pressure high-Tc family or saturate at the current ∼80 K maximum.

The SCLib database is released as an open, versioned resource [6] to support follow-up scoping reviews, validation campaigns, and theory–experiment comparison studies. Subsequent freezes will extend the corpus, refine the extraction pipeline, and progressively close the manual-validation backlog flagged in Section 2.7.

Declarations

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Conflict of interest: The author declares no conflict of interest.

Data availability: All data and code supporting this work are available from the corresponding author upon reasonable request. The SCLib database snapshot `v2026.05.12` and the analysis outputs underlying every figure and table in this manuscript are released through the SCLib repository [6]; the primary literature corpus is drawn from arXiv [7]. The reporting framework follows PRISMA-ScR conventions for scoping reviews [8, 9].

Ethics approval: Not applicable.

Use of large language models: Large language models (Google Gemini 2.5 Flash and Anthropic Claude) were used to assist with material entity recognition during database construction (Section 2.7) and for data extraction assistance during manuscript preparation. All extracted data were validated by the authors. The authors take full responsibility for the accuracy of the reported findings.

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